

Shell history & reverse-search

You will type the same kinds of commands again and again, build, grep a log,
change directories

Recall, edit, re-run

- Arrow up and arrow down step through previous commands
- Ctrl+A and Ctrl+E jump to the start or end of the line
- Ctrl+R starts reverse incremental search
- The history builtin lists numbered entries
- Faster recall means less friction when you iterate on simulators and scripts

Browser lab

INTERACTIVE TOOL

Shell history @ reverse-search

Recall commands with arrows, history, Ctrl+R, and expansions like !! / !n / !\$.
Lab mode — not a full bash.

Starter example: History already has a UART project session. Press `Ctrl+R`, type `make`, then Enter to recall `make test`. Try `history`, `z/s`, and `!!`.

Load starter example

Challenges 0 / 22 cleared

History count: After Load starter, how many history entries? (number)

Answer

Show hintCheckNextIdle

Digital Design and Verification Platform

HomeToolsShell history

Quiz: Ctrl+EQuiz: !!Quiz: !\$esha + History

Lab terminal

Reverse-searchClear screen

Starter session loaded — try Ctrl+R then type `make`

lab\$

History list

Refresh

1 pwd
2 ls -la
3 cd ~/projects/uart_tx
4 make clean
5 make test
6 git status
7 git add src/uart_tx.v
8 git commit -m "fix baud tick"

History: \$0 - last exit 0

Shortcuts & expansions

Key / token	Meaning
<code>↑ / ↓</code>	Step backward / forward through history
<code>Ctrl+R</code>	Reverse incremental search — type a substring; <code>Ctrl+R</code> again = older match
<code>Esc / Ctrl+G</code>	Cancel reverse search
<code>Ctrl+A / Ctrl+E</code>	Cursor to start / end of line
<code>Ctrl+U / Ctrl+K</code>	Kill to start / kill to end
<code>Ctrl+W</code>	Kill word before cursor
<code>history -history N</code>	List all or last N entries
<code>!!</code>	Previous command
<code>!n</code>	Command number n from history
<code>!\$</code>	Last argument of previous command
<code>!prefix</code>	Most recent command starting with prefix

- Focus the prompt and type normally — shortcuts are captured on the input.
- Lab builtins: `help`, `history`, `history -e`, `clear`, `esha`.
- Other commands print a short fake reply and still join history.

Shell history lab

learn_unix — shell history

Real shell practice

```
wsl - module14-shell-history/examples/history

# Track A - real Unix
# real shell session (Track A)

$ set -o history

$ history -s 'pwd'
$ history -s 'ls -la'
$ history -s 'grep ERROR run.log'
$ history -s 'make sim'

$ history
  1  pwd
  2  ls -la
  3  grep ERROR run.log
  4  make sim

$ history | tail -5
  1  pwd
  2  ls -la
  3  grep ERROR run.log
  4  make sim

$ history | grep ERROR
  3  grep ERROR run.log
```

Real shell — history listing

Real shell practice — try these

```
# set -o history — enable history in this shell session
set -o history
```

```
# history -s ... — add a line to history without running it (demo seed)
history -s 'pwd'
```

```
history -s 'ls -la'
```

```
history -s 'grep ERROR run.log'
```

```
history -s 'make sim'
```

```
# history — list numbered past commands
history
```

```
# history | tail -5 — show only the most recent entries
```

Pitfalls to watch

- Bang-number and bang-bang re-execute immediately
- Reverse-search can feel sticky; Ctrl+C cancels cleanly
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On the real shell, list history, search it with grep
- When you are ready, take the short quiz, then continue to aliases and functions